

```
In [4]: import warnings
warnings.filterwarnings('ignore')
# Import pandas for data manipulation
import pandas as pd

# Read the Excel file into a pandas DataFrame
df = pd.read_excel(r"C:\Users\hp\Downloads\Dataset for Data Analytics.xls")

# Display the first few rows to verify it loaded correctly
df.head()
```

```
Out[4]:
```

	OrderID	Date	CustomerID	Product	Quantity	UnitPrice	ShippingAddress
0	ORD200000	2023-01-04	C72649	Monitor	5	570.62	928 Main St
1	ORD200001	2024-08-23	C75739	Phone	2	151.35	823 Main St
2	ORD200002	2024-02-27	C81728	Tablet	5	550.68	512 Main St
3	ORD200003	2023-10-15	C33540	Chair	1	273.19	275 Main St
4	ORD200004	2025-05-08	C81840	Printer	4	626.01	668 Main St

```
In [5]: # Count missing values in each column
print("=== Missing Values Before Cleaning ===\n")
print(df.isnull().sum())
```

=== Missing Values Before Cleaning ===

```
OrderID          0
Date             0
CustomerID       0
Product          0
Quantity         0
UnitPrice        0
ShippingAddress  0
PaymentMethod    0
OrderStatus      0
TrackingNumber   0
ItemsInCart      0
CouponCode       309
ReferralSource   0
TotalPrice       0
dtype: int64
```

```
In [6]: # Count duplicate OrderIDs
duplicate_count = df.duplicated(subset=['OrderID']).sum()
print(f"Number of duplicate OrderIDs: {duplicate_count}")

# Remove duplicates if any (safe to run even if none exist)
df.drop_duplicates(subset=['OrderID'], keep='first', inplace=True)
print(f"Total rows after removing duplicates: {len(df)}")
```

Number of duplicate OrderIDs: 0
Total rows after removing duplicates: 1200

```
In [7]: # Fill empty CouponCode with "None"
df['CouponCode'] = df['CouponCode'].fillna('None')

# Verify no blanks remain in CouponCode
print(f"Missing values in CouponCode after cleaning: {df['CouponCode'].is
```

Missing values in CouponCode after cleaning: 0

```
In [8]: # Convert Date column to proper datetime format
df['Date'] = pd.to_datetime(df['Date'])

# Verify the conversion
print(f"Date column data type: {df['Date'].dtype}")
print(f"Date range: {df['Date'].min()} to {df['Date'].max()}")
```

Date column data type: datetime64[us]

Date range: 2023-01-01 00:00:00 to 2025-06-30 00:00:00

```
In [9]: # Convert numeric columns to proper data types
df['Quantity'] = pd.to_numeric(df['Quantity'])
df['UnitPrice'] = pd.to_numeric(df['UnitPrice'])
df['TotalPrice'] = pd.to_numeric(df['TotalPrice'])
df['ItemsInCart'] = pd.to_numeric(df['ItemsInCart'])

# Quick check of all data types
print("\n=== Data Types After Cleaning ===\n")
print(df.dtypes)
```

=== Data Types After Cleaning ===

OrderID	str
Date	datetime64[us]
CustomerID	str
Product	str
Quantity	int64
UnitPrice	float64
ShippingAddress	str
PaymentMethod	str
OrderStatus	str
TrackingNumber	str
ItemsInCart	int64
CouponCode	str
ReferralSource	str
TotalPrice	float64
dtype:	object

```
In [10]: # Print a complete summary
print("=" * 50)
print("          DATA CLEANING SUMMARY")
print("=" * 50)
print(f"Total rows in cleaned dataset: {df.shape[0]}")
print(f"Total columns: {df.shape[1]}")
print(f"Total missing values after all cleaning: {df.isnull().sum().sum()}")
print(f"Duplicate OrderIDs: {duplicate_count}")
print("\nFirst 3 rows of cleaned data:")
df.head(3)
```

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DATA CLEANING SUMMARY

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Total rows in cleaned dataset: 1200

Total columns: 14

Total missing values after all cleaning: 0

Duplicate OrderIDs: 0

First 3 rows of cleaned data:

Out[10]:

	OrderID	Date	CustomerID	Product	Quantity	UnitPrice	ShippingAddress
0	ORD200000	2023-01-04	C72649	Monitor	5	570.62	928 Main St
1	ORD200001	2024-08-23	C75739	Phone	2	151.35	823 Main St
2	ORD200002	2024-02-27	C81728	Tablet	5	550.68	512 Main St

```
In [12]: # Save cleaned DataFrame to a new Excel file
df.to_excel('Dataset for Analytics CLEANED.xlsx', index=False)

print("Cleaned file saved successfully as 'Dataset for Analytics CLEANED."
```

Cleaned file saved successfully as 'Dataset for Analytics CLEANED.xlsx'

In []: